RAJALAKSHMI ENGINEERING COLLEGE

P Saam 251801276-2025-AIDS

P2

Started on	Thursday, 17 September 2026, 8:21 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:25 AM
Time taken	3 mins 42 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    // Base cases
    if (n == 0 || n == 1 || n == 2) {
        printf("1\n");
        return 0;
    }
    if (n == 3) {
        printf("2\n");
        return 0;
    }

    // Using 64-bit unsigned long long to prevent integer overflow
    unsigned long long ways_n_3 = 1; // ways(1)
    unsigned long long ways_n_2 = 1; // ways(2)
    unsigned long long ways_n_1 = 2; // ways(3)
    unsigned long long current_ways = 0;

    for (int i = 4; i <= n; i++) {
        current_ways = ways_n_1 + ways_n_3; // ways(i) = ways(i-1) + ways(i-3)

        // Shift variables forward for the next iteration
        ways_n_3 = ways_n_2;
        ways_n_2 = ways_n_1;
        ways_n_1 = current_ways;
    }

    // Use %llu format specifier for unsigned long long
    printf("%llu\n", current_ways);
```

```
34
35      return 0;
36  }
37
38
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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